

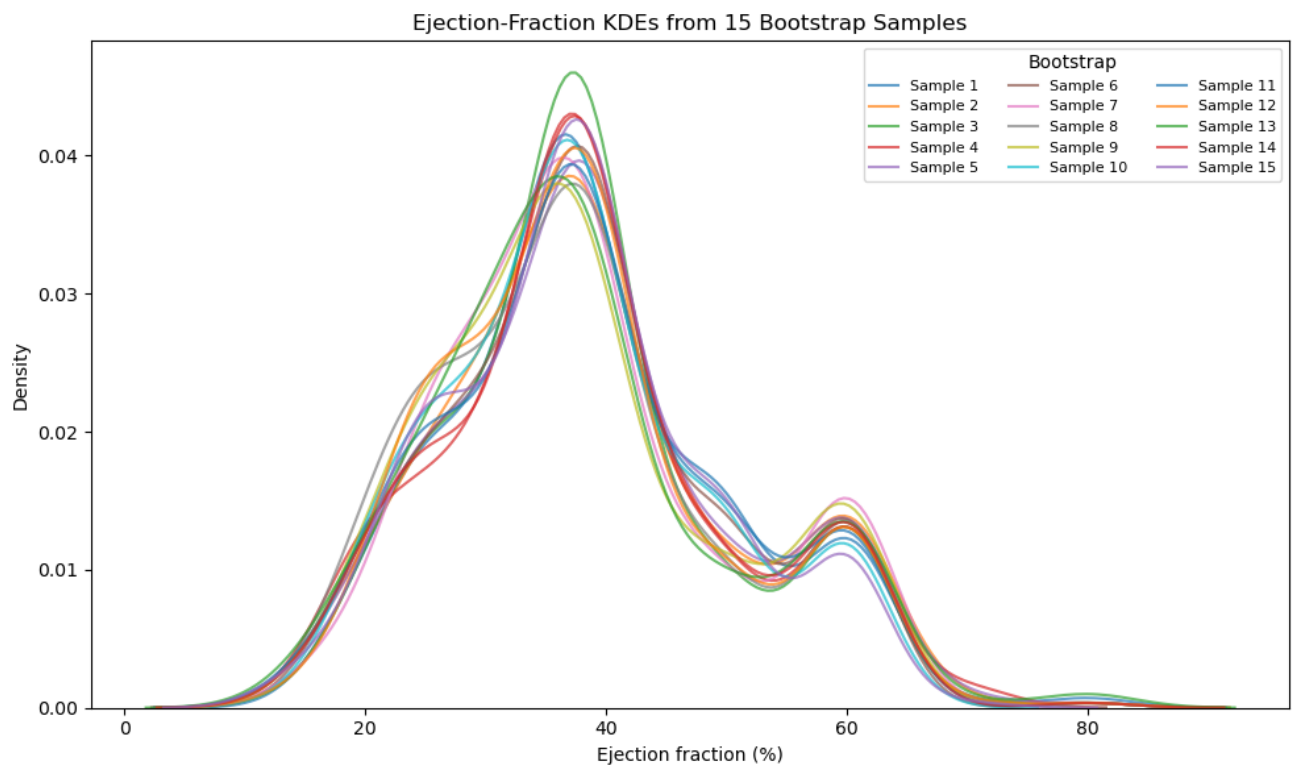
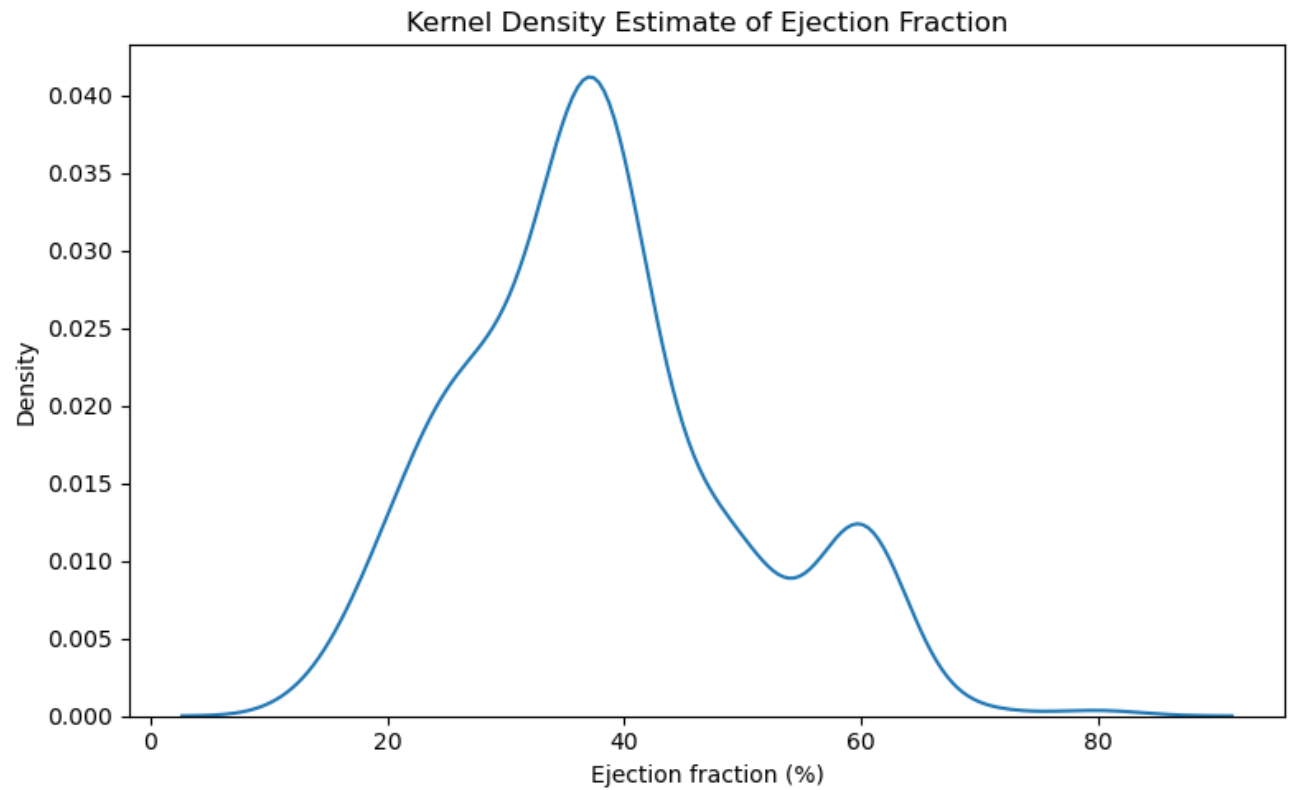
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In [7]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("heart_failure_clinical_records_dataset.csv")

# a. KDE of the original ejection-fraction data
plt.figure(figsize=(8, 5))
sns.kdeplot(data=df, x="ejection_fraction")
plt.title("Kernel Density Estimate of Ejection Fraction")
plt.xlabel("Ejection fraction (%)")
plt.ylabel("Density")
plt.tight_layout()
plt.show()

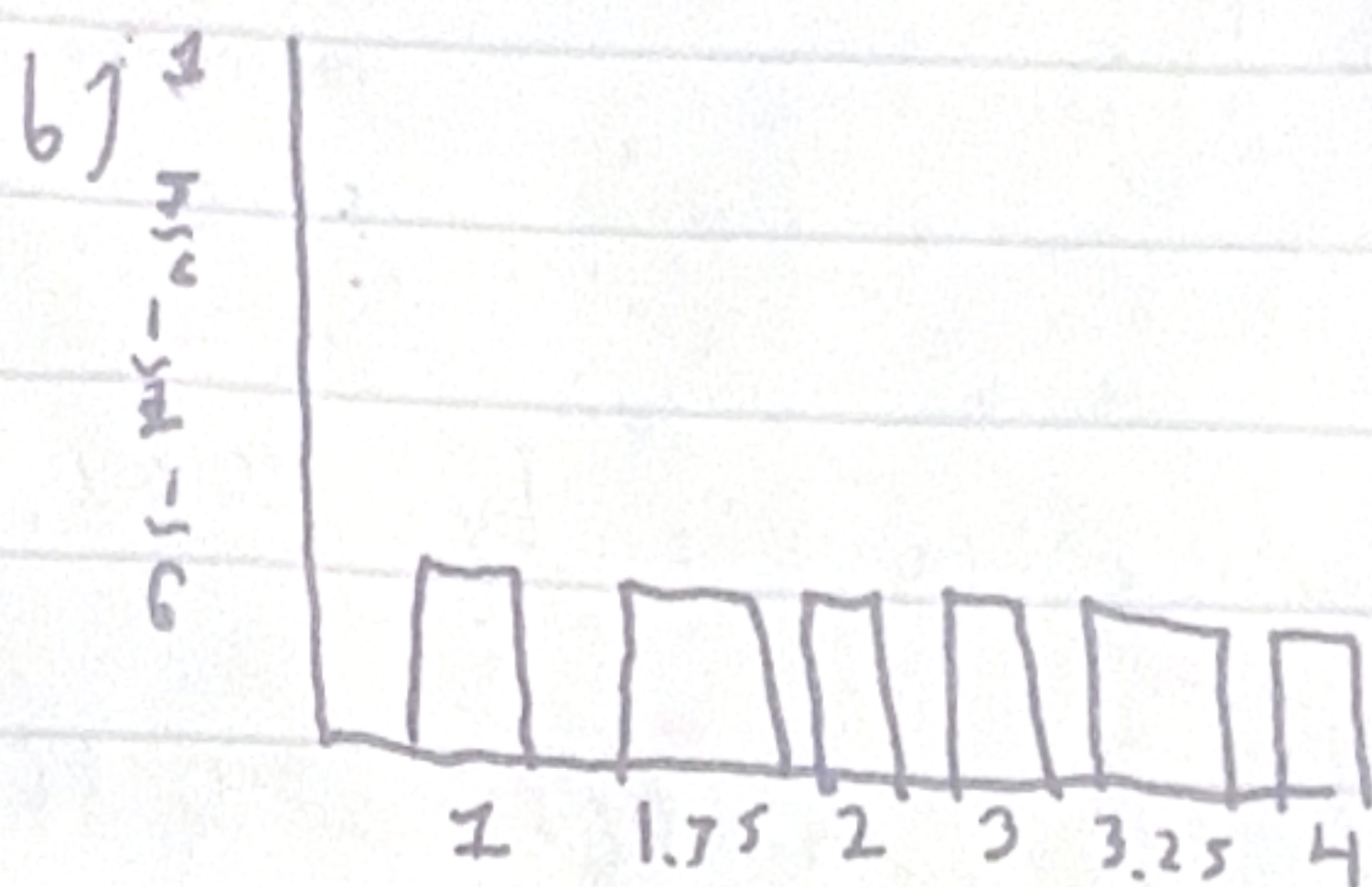
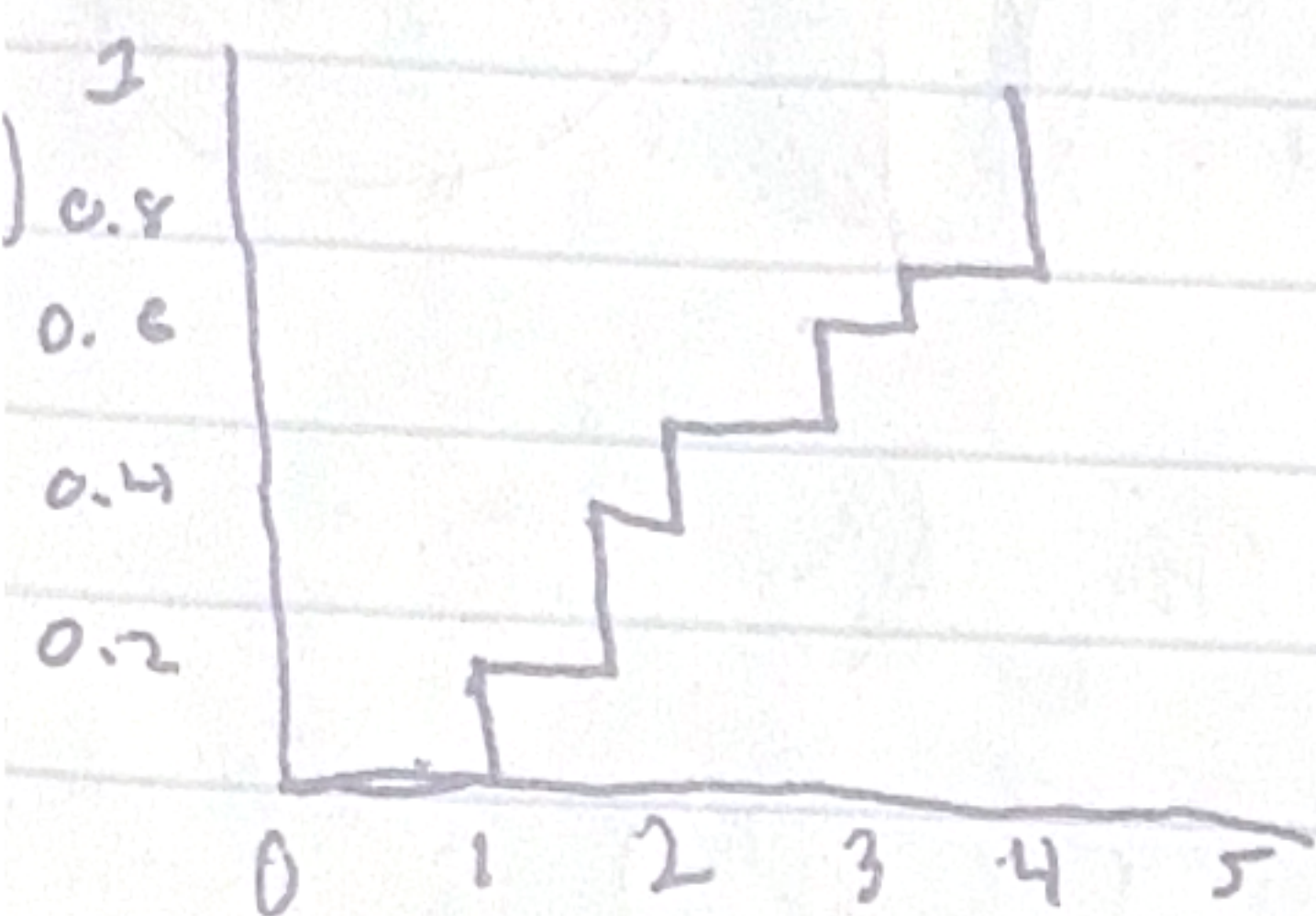
# b. KDEs from 15 bootstrap samples of the same size
plt.figure(figsize=(10, 6))
for sample_number in range(1, 16):
    bootstrap_sample = df.sample(
        frac=1.0,
        replace=True,
        random_state=sample_number,
    )
    sns.kdeplot(
        data=bootstrap_sample,
        x="ejection_fraction",
        label=f"Sample {sample_number}",
        alpha=0.7,
    )

plt.title("Ejection-Fraction KDEs from 15 Bootstrap Samples")
plt.xlabel("Ejection fraction (%)")
plt.ylabel("Density")
plt.legend(title="Bootstrap", ncol=3, fontsize=8)
plt.tight_layout()
plt.show()
```



4c) Overall the original KDE appears to be very reliable. Both tails (0-20) and 70+ look very reliable. Immediately after the peak (40-45) also looks very reliable. The largest discrepancies are 25-35, the peak (35-40), and (55-65)

Staden Rothschild



it is uniform bc with Δh of 0.2 there are no overlaps.

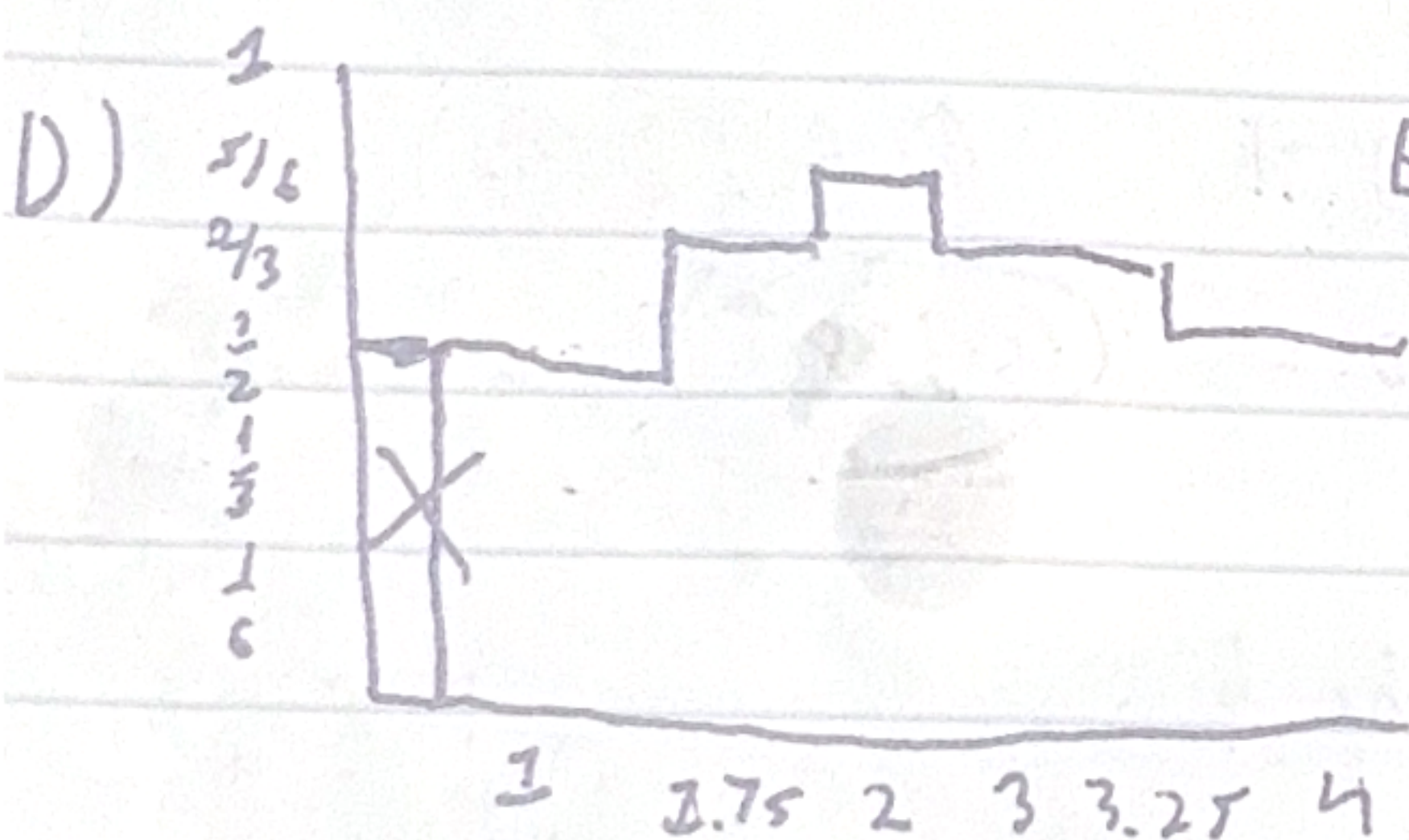
$$h = 1.94 \times 50 \times n^{-0.2}$$

$$1.94 \times 1.02 \times 6^{-0.2} = \boxed{1.3}$$

$$n = (1 + 1.75 + 2 + 3 + 3.25 + 4) / 6 = 2.5$$

$$SD = \sqrt{\frac{1}{6} [(1-2.5)^2 + (1.75-2.5)^2 + (2-2.5)^2 + (3-2.5)^2 + (3.25-2.5)^2 + (4-2.5)^2]}$$

$$2.25 + 0.5625 + 0.25 + 0.25 + 0.5625 + 2.25 / 6 = \sqrt{1.02} \approx 1.02$$



E) In D none of the kernel overlap with one another with the kernel bandwidth. In U, they will overlap with at least one point so the densities will be higher than $1/n$.

$$2. \hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x-x_i}{h}\right)$$

$$K(u) = \begin{cases} \frac{1}{2} & |u| \leq 1 \\ 0 & |u| > 1 \end{cases}$$

three kernel contribute: -0.5, 0, 0.5

$$\hat{f}_{0.6}(0) = \frac{1}{5(0.6)} \left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2} \right) \approx \frac{1.5}{3} = \boxed{0.5}$$

$$b) E[\hat{f}_h(u)] = \frac{1}{2h} P(|X| \leq h) \quad E[\hat{f}_{0.6}(0)] = \frac{1}{1.2} P(-0.6 \leq X \leq 0.6)$$

$$P(-0.6 \leq X \leq 0.6) = \Phi(0.6) - \Phi(-0.6) \quad \Phi(0.6) = 0.73$$

$$P(-0.6 \leq X \leq 0.6) \approx 2(0.73) - 1 = 0.46$$

$$E[\hat{f}_{0.6}(0)] = \frac{0.46}{1.2} = \boxed{0.38}$$

$$2c) f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$f(0) = \frac{1}{\sqrt{2\pi}} \approx 0.3989$$

$$0.3989$$

d) No, they are off by 0.02. The bias is

$$E[\hat{f}_h(0)] - f(0) = 0.38 - 0.4 = -0.02$$

$E[\hat{f}_h(0)] \approx 0.38$ is what we expect the KDE to equal on average across many samples.

$$e) E[\hat{f}_{0.1}(0)] = \frac{1}{2(0.1)} P(-0.1 \leq X \leq 0.1)$$

$$P(-0.1 \leq X \leq 0.1) = 2\Phi(0.1) - 1 \quad \Phi(0.1) \approx 0.54$$

$$P(-0.1 \leq X \leq 0.1) \approx 0.0796 / 0.2 = 0.3983$$

As h shrinks, so does the bias.

f) Yes, since as $h \rightarrow 0$, the bias shrinks to 0 as well.

g) The tradeoff between bias and variance. A small h will have less bias, but will have more variance by capturing more noise in the data. A higher h will be smoother and capture less noise, but may also have a higher bias.

$$3a) \hat{f}_h(x) = \frac{1}{5(1.5)} \left(\frac{1}{2} \times \# \text{ observations within } 1.5 \right) = \frac{\text{number of observations}}{15}$$

$$\begin{aligned} 6.0 &= \frac{3}{15} = 0.2 \\ 6.5 &= \frac{4}{15} = 0.27 \\ 7.0 &= \frac{3}{15} = 0.2 \\ 8.0 &= \frac{3}{15} = 0.2 \\ 8.5 &= \frac{4}{15} = 0.27 \end{aligned}$$

$$\begin{aligned} 3b) \hat{f}_h(x) &= \frac{1}{5(1.5)} \sum_{i=1}^5 \phi\left(\frac{x - x_i}{h}\right) \\ 6.0 &\rightarrow 0.1512 \\ 6.5 &\rightarrow 0.1687 \\ 7 &\rightarrow 0.1780 \\ 8 &\rightarrow 0.1674 \\ 8.5 &\rightarrow 0.1506 \end{aligned}$$

3c) The biggest jump is 0.07 between $(-6, 6.5)$, $(6.5, 7)$ and $(8, 8.5]$. The uniform kernel has a hard cutoff. A point contributes to a fixed amount inside the bandwidth and zero outside. Values can jump suddenly when a data point enters or leaves the window. The gaussian kernel has no cutoff and its weights decrease gradually with distance, so its KDE changes smoothly.

3d) The uniform kernel is simple and computationally convenient, but its hard cutoff causes abrupt jumps when observations enter or leave the bandwidth. The gaussian kernel assigns weights that decrease smoothly with distance, producing a continuous density estimate.